

CSE321 – Lab Session 1 Report

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1 Introduction

This report reflects on the ER modelling exercise I completed during Lab Session #1 of CSE 321 Database Systems. The task required constructing an Entity-Relationship (ER) diagram in Chen’s notation from a written specification describing the data requirements of a university system. The specification covered four groups of people (professors, staff, students, and stakeholders), two types of staff, two types of students, a laboratory system with equipment, courses, supervision, and stakeholder comments.

The purpose of this report is to critically examine my original in-class submission: identifying what I modelled correctly, where my reasoning was flawed, what assumptions I made during the session and whether they were justified, and how I revised the design after carefully re-reading the specification. The aim is not simply to present a corrected diagram, but to demonstrate honest understanding of where my conceptual model diverged from the requirements and why.

The ER modelling notation used throughout this report follows the Chen notation as described in [1] as was required. The digital diagrams I drew using online tool called Visual Paradigm Online [2].

2 Review of the Original Design

2.1 Original ER Diagram

Figure 1 shows the ER diagram I submitted during the in-class lab session.

2.2 Reasoning Behind the Original Design

My original design was built around a UNIVERSITY entity at the centre of the diagram. I connected it to PROFESSOR, STUDENT, STAFF, and STAKEHOLDERS through a single *depend on* relationship. This decision came from the opening sentence of the specification – “there are four different groups of people on whom University is most dependent” – which I read as a modelling instruction rather than as contextual motivation. I also connected the shared person attributes (person_ID, name, email, address, phone, school, zip code, nation) directly to STAKEHOLDERS, as that was the entity I happened to be working on when I read those attributes in the specification.

For courses, I drew a *teach* relationship between PROFESSOR and COURSE and a *take* relationship between STUDENT and COURSE, with course_ID, name, date, and credits as attributes. I noted that the specification used the term “Date_Time” but wrote it simply as *date* under time pressure.

For the student hierarchy, I used an ISA triangle connecting STUDENT to GRADUATE and UNDERGRADUATE. I then introduced a further ISA level below GRADUATE, creating a RESEARCH GRADUATE subtype. *This was a deliberate assumption I documented on my submitted paper:* I interpreted the specification phrase “graduates who

do research” as defining a distinct subset of graduates, and so I modelled them as a separate named subclass. I connected the *supervise* relationship (with *research topic*) and the research lab assignment to RESEARCH GRADUATE rather than to GRADUATE directly.

For undergraduate students, I drew a *conduct experiment* relationship connecting UNDERGRADUATE to TEACHING LAB. For laboratories I drew a LAB entity with attributes including name, department number, location, capacity, and a date field, along with ISA specialisations to TEACHING LAB and RESEARCH LAB. I used double borders for LAB, *has/contains*, and EQUIPMENT, recognising that equipment is a weak entity whose identity depends on its laboratory. I included ID (underlined), name, Date_Purchased, and Model_No as equipment attributes.

For staff I drew an ISA hierarchy from STAFF to ADMIN STAFF and TECH STAFF, and a further ISA from ADMIN STAFF to ADMIN MANAGER. I connected TECH STAFF to LAB via an *assigned* relationship. I also drew a *manage* relationship between STAFF and a DAILY OPERATIONS entity, based on the specification’s phrase “help to manage the schools’ daily operations.”

For stakeholders I drew a *provide comment* relationship and connected it to a COMMENT entity with attributes *time stamp*, *topic*, and *date*. For the domain I drew STAKEHOLDERS connected via *belong to* to a DOMAIN entity.

I did not include any total participation constraints (double lines) in my diagram, and I did not model the *Hires* relationship between PROFESSOR and ADMIN MANAGER.

Assumptions Made During the Lab Session.

- The phrase “the university depends on” was modelling language, so I drew a UNIVERSITY entity connected to all four groups via *depend on*.
- The shared person attributes (person_ID, name, etc.) were placed on STAKEHOLDERS only, as I did not recognise the need for a shared supertype.
- **Graduates who do research are a subclass of graduate students**, which I named RESEARCH GRADUATE and modelled as a further ISA specialisation below GRADUATE.
- Equipment is a weak entity dependent on Laboratory – I deliberately used double borders for Equipment, *has/contains*, and Laboratory.
- “Daily operations” referred to a data entity that staff manage.
- Domain is a separate entity rather than a simple attribute of Stakeholder.
- Total participation was not required to be shown explicitly.
- The *Hires* relationship between Professor and Admin Manager was not needed.

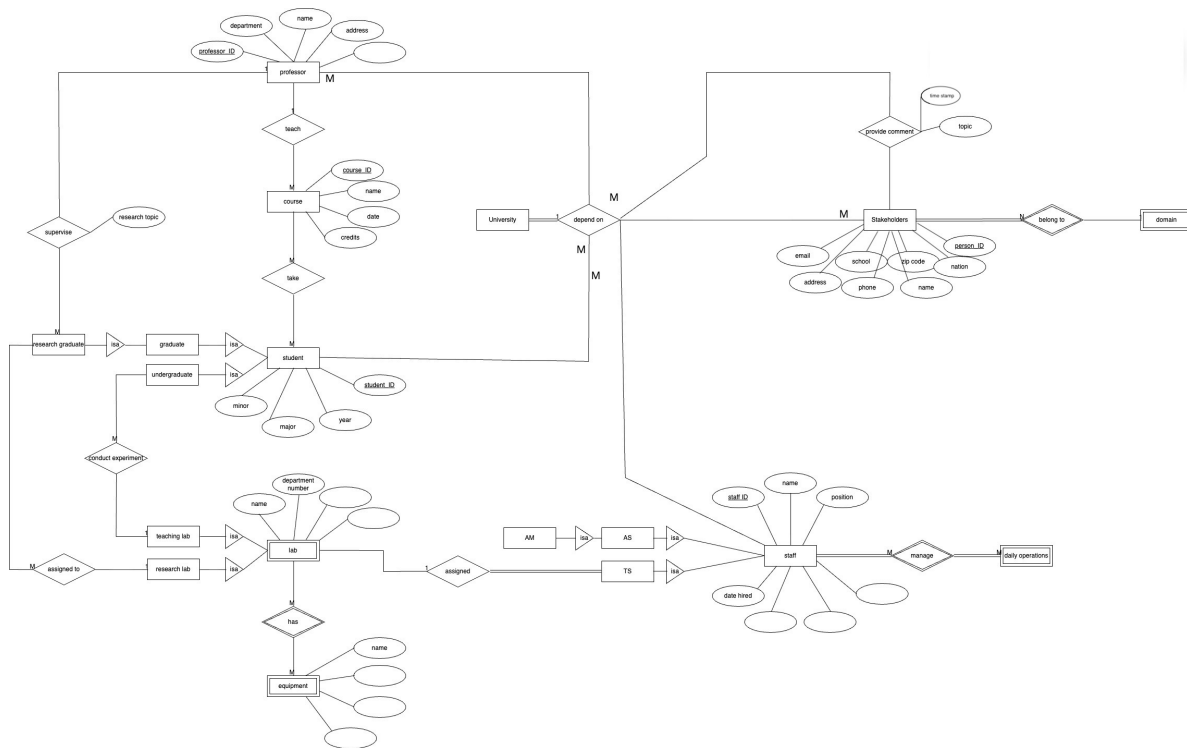


Figure 1: Original ER Diagram submitted during the in-class lab session.

3 Revision Details

3.1 Revised ER Diagram

Figure 2 shows my revised ER diagram, constructed after carefully re-reading the specification. Each revision is explained in the subsections below.

Revision 1: Removal of UNIVERSITY entity; introduction of PERSON supertype

What was wrong: I drew a UNIVERSITY entity connected to all four person groups via a *depend on* relationship. The shared person attributes appeared only on STAKEHOLDERS, not on a shared structure accessible to all four groups.

Why it was conceptually incorrect: The specification does not define a UNIVERSITY entity — the phrase “on whom University is most dependent” is motivational language, not a data requirement. No attributes or independent relationships are assigned to University in the specification, so it has no legitimate place in the diagram. More critically, the specification explicitly states that Person_ID, Name, Zipcode, Address, Email, Nation, Phone, and Schools are *shared* by all four person groups. In ER modelling, attributes shared across multiple entity types must be factored into a *supertype*, with the individual groups as subtypes connected via an ISA relationship. Placing them only on STAKEHOLDERS and having four disconnected flat entities means the shared structure is entirely missing from the model.

How the revision fixes it: As shown in Figure 3, the UNIVERSITY entity and *depend on* relationship are removed entirely. A

PERSON supertype is introduced with Person_ID as the primary key and all eight shared attributes attached to it. The four role entities — PROFESSOR, STAFF, STUDENT, and STAKEHOLDER — become ISA subtypes of PERSON, each defining only their own unique attributes.

Original: UNIVERSITY entity + *depend on* relationship; shared attributes placed only on STAKEHOLDERS.

Revised: PERSON supertype with all 8 shared attributes and Person_ID as key; four ISA subtypes each defining unique attributes only.

Revision 2: Removal of RESEARCH GRADUATE subtype

What was wrong: I introduced a three-level student hierarchy: STUDENT → ISA → {GRADUATE, UNDERGRADUATE}, and then a further ISA below GRADUATE producing a RESEARCH GRADUATE subtype. The *supervise* relationship and the research lab assignment were then attached to RESEARCH GRADUATE rather than to GRADUATE.

The assumption I made: I noted on my submitted paper during the lab session that I assumed “graduates who do research” to be a distinct subclass of graduate students, which I named RESEARCH GRADUATE. My reasoning was that the specification’s phrase “graduates *who do research*” implied a qualifying condition — not all graduates, only a subset — and that a qualifying condition on a subtype warrants a new ISA level.

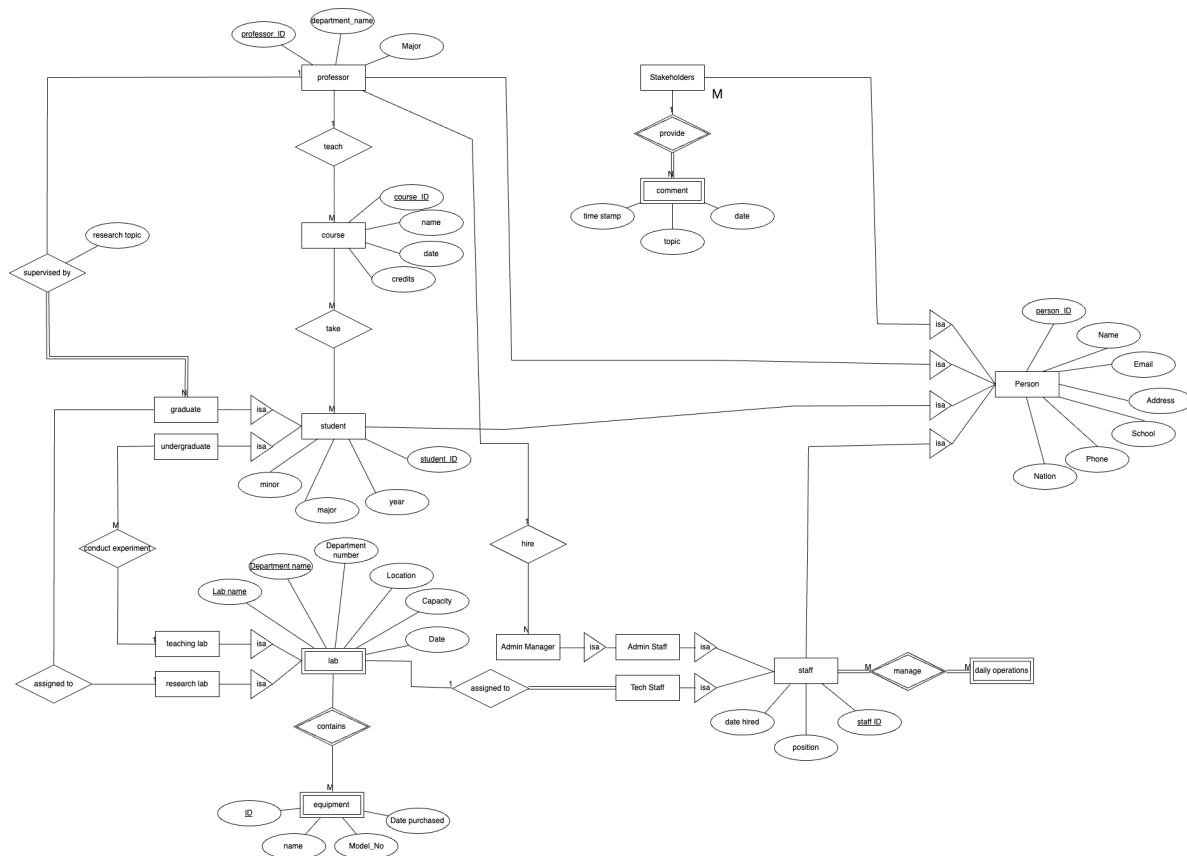


Figure 2: Revised ER Diagram in Chen's notation.

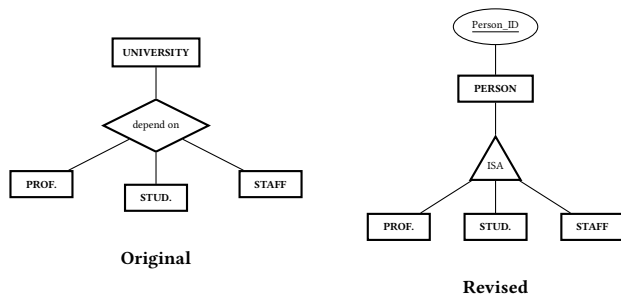


Figure 3: Revision 1: UNIVERSITY removed; PERSON super-type introduced.

Why the assumption was wrong: Although the reasoning felt logical at the time, it is a case of over-modelling. In ER modelling, a subtype is only justified when it has *unique attributes* or *participates in relationships that the parent type does not*. The RESEARCH GRADUATE subtype in my diagram had neither: it carried no new attributes of its own, and all of its relationships (supervised by a professor, assigned to a research lab) apply to *all* graduate students according to the specification. The phrase “graduates who do research” is a description of what graduate students generally do in

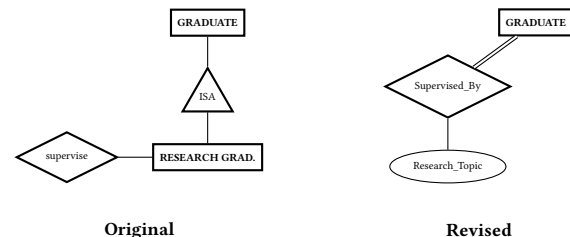


Figure 4: Revision 2: RESEARCH GRADUATE removed; Supervised_By connects GRADUATE directly to PROFESSOR.

this university system, not a definition of a further category. The specification defines exactly two student subtypes — undergraduate and graduate — and does not define any third category.

Furthermore, placing the supervision and research lab relationships on RESEARCH GRADUATE rather than on GRADUATE creates a logical inconsistency: a graduate student who is not a RESEARCH GRADUATE in my model would have no supervision relationship at all, which contradicts the specification's intent. The correct design attaches these relationships directly to GRADUATE with total participation, meaning every graduate must be supervised by exactly one professor and must be assigned to at least one research lab.

How the revision fixes it: As shown in Figure 4, RESEARCH GRADUATE and its ISA triangle are removed. The *Supervised_By* relationship is attached directly between GRADUATE and PROFESSOR, with *Research_Topic* as a relationship attribute and a double line from the GRADUATE side (total participation). The *Assigned_To* research lab relationship is similarly attached directly to GRADUATE.

Original: STUDENT → GRADUATE → RESEARCH GRADUATE (three levels); supervision and lab assignment on RESEARCH GRADUATE.

Revised: STUDENT → GRADUATE / UNDERGRAD (two levels only); Supervised_By and Assigned_To connect directly to GRADUATE with total participation.

Revision 3: DAILY OPERATIONS is not an entity

What was wrong: I drew DAILY OPERATIONS as a separate rectangular entity connected to STAFF via a *manage* relationship (M:M).

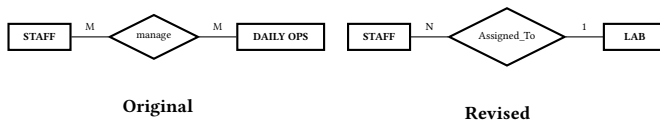


Figure 5: Revision 3: DAILY OPERATIONS removed; Tech Staff's Assigned_To Lab correctly placed.

Why it was conceptually incorrect: The phrase “they help to manage the schools’ daily operations” appears in the specification as a description of staff’s *purpose*, not as a definition of a data object. DAILY OPERATIONS has no attributes, no key, and no independent relationships of its own anywhere in the specification. An entity in an ER diagram must represent something for which data needs to be stored; a narrative phrase describing what staff generally do does not meet this criterion. Treating it as an entity creates a spurious object that cannot be meaningfully populated in a database.

How the revision fixes it: DAILY OPERATIONS and the *manage* relationship are removed entirely. The relevant relationship for staff in the specification is that TECH STAFF is assigned to at most one laboratory — this is correctly represented by the *Assigned_To* relationship between TECH STAFF and LABORATORY (N:1).

Original: DAILY OPERATIONS drawn as an entity; STAFF connected via *manage* (M:M).

Revised: DAILY OPERATIONS and *manage* removed entirely; Tech Staff’s Assigned_To Lab (N:1) is the correct staff–lab relationship.

Revision 4: DOMAIN is an attribute, not an entity

What was wrong: I drew DOMAIN as a separate rectangular entity connected to STAKEHOLDERS via a *belong to* relationship (N:1).

Why it was conceptually incorrect: The specification states that stakeholders “belong to a particular Domain (e.g., funding agency, government, industry partners, and public).” The parenthetical examples are categorical values, not independent data objects. DOMAIN has no attributes or relationships of its own in the specification. In ER modelling, when something is simply a categorical

value that describes an entity — with no key, no sub-structure, and no relationships of its own — it is correctly modelled as a plain single-valued *attribute* of that entity, not as a separate entity with a relationship.

How the revision fixes it: DOMAIN becomes a plain attribute of STAKEHOLDER. The *belong to* relationship is removed.

Original: DOMAIN drawn as a separate entity; STAKEHOLDERS connected via *belong to* (N:1).

Revised: Domain is a simple attribute of STAKEHOLDER; *belong to* relationship removed.

Revision 5: Missing total participation constraints

What was wrong: My original diagram contained no double lines (total participation) anywhere, despite the specification explicitly stating constraints that require them. In particular, there were no double lines on the graduate side of the *supervise* or *assigned to* relationships, and no double line on the COMMENT side of *provide*.

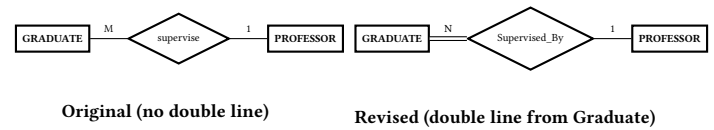


Figure 6: Revision 5: Total participation added on GRADUATE side of Supervised_By.

Why it was conceptually incorrect: Total participation (double line) encodes the constraint that every instance of an entity *must* participate in the relationship — it is not cosmetic. The specification uses precise language that maps directly to these constraints: “supervised by *exactly one* professor” means total participation (cardinality 1) from the graduate side; “assigned to *at least one* research laboratory” means total participation (cardinality M) from the graduate side. Omitting these double lines produces a diagram that is structurally present but semantically incomplete — it cannot enforce that every graduate has a supervisor or a research lab assignment.

How the revision fixes it: As shown in Figure 6, double lines (total participation) are applied from GRADUATE to *Supervised_By* (every graduate must be supervised by exactly one professor) and from GRADUATE to *Assigned_To* (every graduate must be assigned to at least one research lab). Total participation is also applied from COMMENT to *Provides* (a comment cannot exist without a stakeholder).

Original: No total participation shown anywhere; participation constraints from the specification not represented.

Revised: Double lines on GRADUATE → Supervised_By; GRADUATE → Assigned_To (Research Lab); COMMENT → Provides.

Revision 6: Missing Hires relationship between Professor and Admin Manager

What was wrong: My original diagram did not contain a relationship between PROFESSOR and ADMIN MANAGER. The staff hierarchy (Staff → Admin Staff → Admin Manager) was present, but the connection to Professor was absent.

Why it was conceptually incorrect: The specification explicitly states: “Each professor can hire several administrative managers.” This is a direct, unambiguous data requirement for a relationship with a defined cardinality (1 professor to N admin managers). Omitting it means the diagram fails to represent a real and specified data requirement. The relationship also implicitly carries a participation constraint: an Admin Manager exists only because a professor hires them, suggesting total participation from the Admin Manager side.

How the revision fixes it: A *Hires* relationship is added between PROFESSOR and ADMIN MANAGER with cardinality 1:N (one professor can hire multiple admin managers). This correctly models the specified requirement.

Original: No relationship between PROFESSOR and ADMIN MANAGER; *Hires* entirely absent.

Revised: *Hires* relationship added between PROFESSOR (1) and ADMIN MANAGER (N).

4 Conclusion

Reviewing my own diagram against the specification after the lab session made the nature of my mistakes much clearer than I expected. The errors fell into two distinct categories: misreading narrative language as modelling instructions, and gaps in applying participation constraints under time pressure.

The most consequential structural error was introducing a UNIVERSITY entity. I now understand that a specification must be read for *data objects, their properties, and their relationships* — not for narrative framing. The phrase “the university depends on” is an explanation of why the four groups matter, not a definition of an entity. Similarly, DAILY OPERATIONS was drawn from a descriptive sentence about staff purpose, not from any defined data object in the specification. In future, before drawing any entity I will ask: does this thing have a key, attributes, and relationships of its own defined in the specification? If not, it is not an entity.

The RESEARCH GRADUATE mistake taught me a different lesson: the importance of restraint in specialisation. My assumption — that “graduates who do research” implied a distinct subclass — felt justified at the time because of the qualifying language. However, I now understand that a subtype requires either unique attributes or unique relationships that the parent cannot have. Since all graduate students in this system are supervised and assigned to research labs, the qualification describes the general condition of being a graduate student, not a further category. Over-modelling by adding unnecessary ISA levels creates complexity without adding information.

The missing total participation constraints revealed that I understood the notation conceptually but did not apply it systematically. The specification’s use of “exactly one” and “at least one” are direct encodings of total participation, and I will treat these phrases as explicit triggers for double lines in future diagrams.

Finally, the missing *Hires* relationship was a straightforward reading error under time pressure. It reinforced the importance of checking every stated relationship in the specification against the diagram before submitting, rather than moving on once the main structure is in place.

References

- [1] Ramez Elmasri and Shamkant B. Navathe. 2015. *Fundamentals of Database Systems* (7th ed.). Pearson.
- [2] Visual Paradigm International. 2025. Visual Paradigm Online Diagramming Tool. <https://online.visual-paradigm.com>. Accessed: April 2026.